

Fall 2013

Problem 1 (Fa13). The set of pairs of positive real numbers (x, y) with $x^y = y^x$ is a union of two smooth curves. Find the point where they intersect.

Problem 2 (Fa13). Suppose that x is a smooth real valued function of the real number t , satisfying $x'(t) \leq b(t)x(t)$ for some continuous function b . Prove that if $s \leq t$ then $x(t) \leq x(s) \exp \int_s^t b(t) dt$.

Problem 3 (Fa87, Sp92, Fa97, Sp00, Fa13, Sp19). Define a sequence of real numbers (x_n) by

$$x_0 = 1 \quad x_{n+1} = \frac{1}{2 + x_n} \quad \text{for } n \geq 0$$

Show that (x_n) converges and evaluate its limit.

Problem 4 (Fa86, Fa13, Fa19, Fa23). Show that the polynomial $p(z) = z^5 - 6z + 3$ has five distinct complex zeros, of which exactly three (and not five) are real.

Problem 5 (Sp88, Fa13). For $a > 1$ and $n = 0, 1, 2, \dots$, evaluate the integrals

$$C_n(a) = \int_{-\pi}^{\pi} \frac{\cos n\theta}{a - \cos \theta} d\theta \quad S_n(a) = \int_{-\pi}^{\pi} \frac{\sin n\theta}{a - \cos \theta} d\theta$$

Problem 6 (Fa13). Let V be the complex vector space of complex 2×2 matrices X . Find all quadratic forms Q on V such that $Q(X) = Q(AXA^{-1})$ for any complex invertible 2×2 matrix A .

Problem 7 (Sp80, Sp10, Fa13, Sp18). Let A and B be $n \times n$ complex matrices. Prove or disprove each of the following statements:

1. If A and B are diagonalizable, so is $A + B$.
2. If A and B are diagonalizable, so is AB .
3. If $A^2 = A$, then A is diagonalizable.
4. If A is invertible and A^2 is diagonalizable, then A is diagonalizable.
5. If AB is diagonalizable and invertible then so is BA .

Problem 8 (Fa94, Fa13, Sp24). Let R be a ring with identity, and let u be an element of R with a right inverse. Prove that the following conditions on u are equivalent:

1. u has more than one right inverse;
2. u is a zero divisor;
3. u is not a unit.

Problem 9 (Fa13). 1. Give an example of a group G such that the center of G modulo its own center is nontrivial.

2. Give an example of a group H such that the groups in its derived series $H \supset H' \supset H'' \supset H'''$ are all distinct.

Problem 10 (Su80, Sp97, Fa11, Fa13, Fa17, Sp21, Fa25). For each $(a, b, c) \in \mathbb{R}^3$, consider the series

$$\sum_{n=3}^{\infty} \frac{a^n}{n^b \log^c n}$$

Determine the values of (a, b, c) for which the series converges absolutely; converges but not absolutely; diverges.

Problem 11 (Fa13). Let X be a metric space. Prove that X is connected if and only if the image of every continuous function $f : X \rightarrow \mathbb{R}$ is an interval.

Problem 12 (Fa13). Let $f : (0, 1) \rightarrow \mathbb{R}$ be a continuous function with $\int_0^1 |f(t)| dt < \infty$. Define $g : (0, 1) \rightarrow \mathbb{R}$ by

$$g(x) = \int_x^1 \frac{f(t)}{t} dt$$

Show that $\int_0^1 |g(x)| dx < \infty$.

Problem 13 (Su84, Fa13). Evaluate $\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 4x + 20} dx$.

Problem 14 (Fa13). Let $U \subset \mathbb{C}$ be a bounded open set containing 0, and let $f : U \rightarrow \mathbb{C}$ be an analytic function, whose Taylor series at 0 is

$$f(z) = z + a_2 z^2 + a_3 z^3 + \dots$$

Prove that $a_2 = 0$.

Problem 15 (Fa13). Is it possible to find two real 2×2 matrices A, B such that $A^2 = B^2 = I$, but AB has eigenvalues 2 and $1/2$?

Problem 16 (Fa13). Suppose that A is an $m \times n$ complex matrix and B is an $n \times m$ complex matrix, and write I_m for the $m \times m$ identity matrix. Show that if $I_m - AB$ is invertible then so is $I_n - BA$.

Problem 17 (Fa13). Prove that if n is coprime to $N = 561$ then $n^{N-1} \equiv 1 \pmod{N}$.

Problem 18 (Fa13). How many irreducible polynomials of degree exactly 6 are there over the finite field with 3 elements?